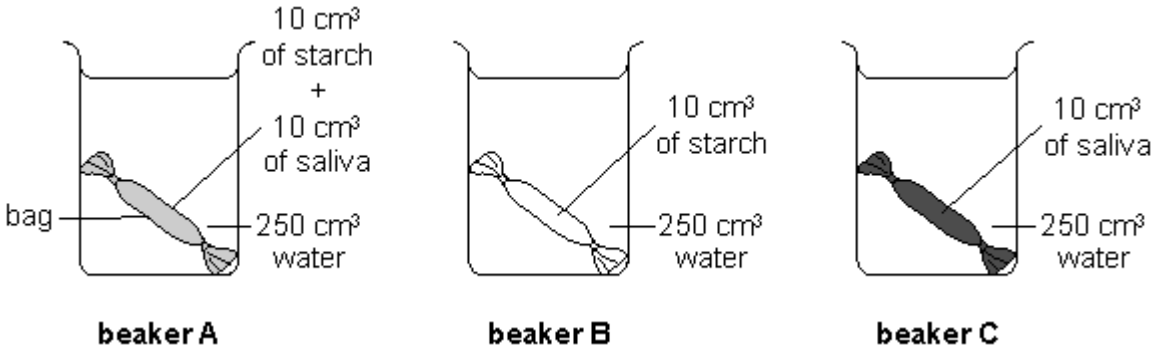
**Q1.**

Sally investigated how the human body digests and absorbs starch.

She used saliva to digest the starch.

To model digestion she used special bags made from a semi-permeable membrane. These bags have lots of very small holes.

Sally sets up the equipment as shown below. There is one special bag in each beaker.




**beaker A** **beaker B** **beaker C**

She keeps the water in the beakers at 37ºC.
After 20 minutes, Sally tested the contents of each beaker and bag for starch and sugar.
The table below shows Sally's results.

| | Was starch found in the bag? | Was sugar found in the bag? | Was starch found in the water? | Was sugar found in the water? |
|---|---|---|---|---|
| beaker A | ✓ | ✓ | ✗ | ✓ |
| beaker B | ✓ | ✗ | ✗ | ✗ |
| beaker C | ✗ | ✗ | ✗ | ✗ |

(a) Suggest why Sally kept the water at 37ºC.

.....

1 mark

(b) (i) Explain why sugar was found in the bag in beaker A.

.....

1 mark

(ii) Starch was **not** found in the **water** outside the bag in any beaker.
Suggest why.

.....

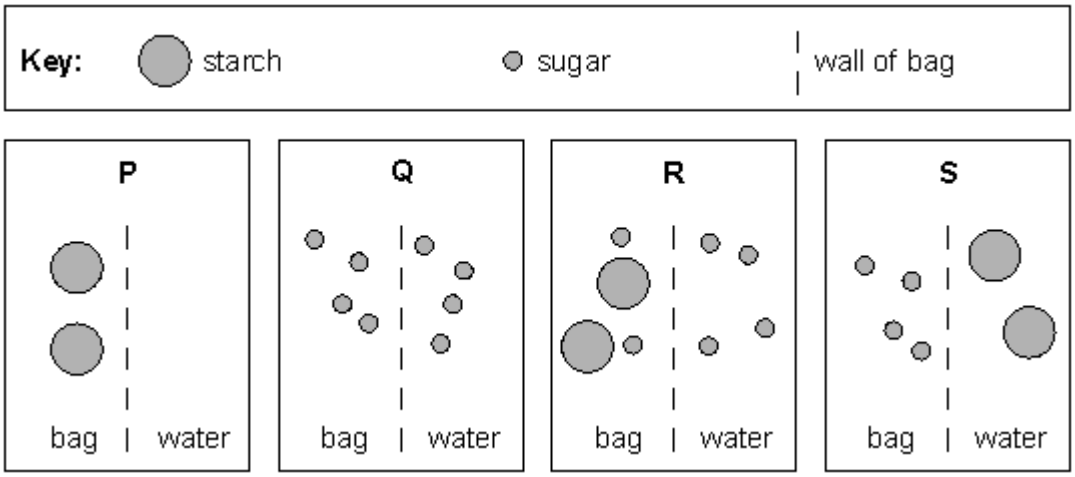
1 mark

(c) Why did Sally set up beaker C? Tick the correct box.

| | | | |
|---|---|---|---|
| for a fair test | ☐ | for accuracy | ☐ |
| for reliability | ☐ | for a control | ☐ |

1 mark

(d) Sally used diagrams to show what happened in her investigation.




Use the diagrams above to answer the following questions.

(i) Which diagram shows the **results** of beaker **B**? Write the letter.

.....

1 mark

(ii) Which diagram shows the **results** of beaker **A**? Write the letter.

.....

1 mark

(e) What does saliva contain that causes starch to change in beaker A?

.....

1 mark

(f) Sally chewed a piece of bread for 5 minutes without swallowing.
What would she notice about the taste of the bread after chewing for 5 minutes?
Use Sally's results to help you.

.....

1 mark
maximum 8 marks

## Q1.

(a) • (37 °C is) body temperature

*accept 'so the saliva **or** enzymes would work'*
*accept 'it is a good **or** optimum temperature for digestion'*
*'to make it a fair test' is insufficient*
*'so they are all the same' is insufficient*

**1 (L5)**

(b) (i) • the starch is broken down **or** digested

*'there is a reaction between starch and saliva' is insufficient*

**1 (L6)**

(ii) any **one** from

• starch could not pass through the bag

*accept 'starch could not get through the holes'*
*'the bag is semi-permeable' is insufficient*

• starch is too big

*'the bag holds it in' is insufficient*

**1 (L6)**

(c) • for a control ✓

*if more than one box is ticked, award no mark*

**1 (L6)**

(d) (i) • P

*if more than one letter is given, award no mark*

**1 (L6)**

(ii) • R

*if more than one letter is given, award no mark*

**1 (L6)**

(e) • enzymes

*accept 'amylase' **or** 'carbohydrase'*

**1 (L6)**

(f) any **one** from

• sweeter **or** sugary

*accept 'sugar' **or** 'sweet'*

• it tastes of sugar (L6)

**1**

**[8]**